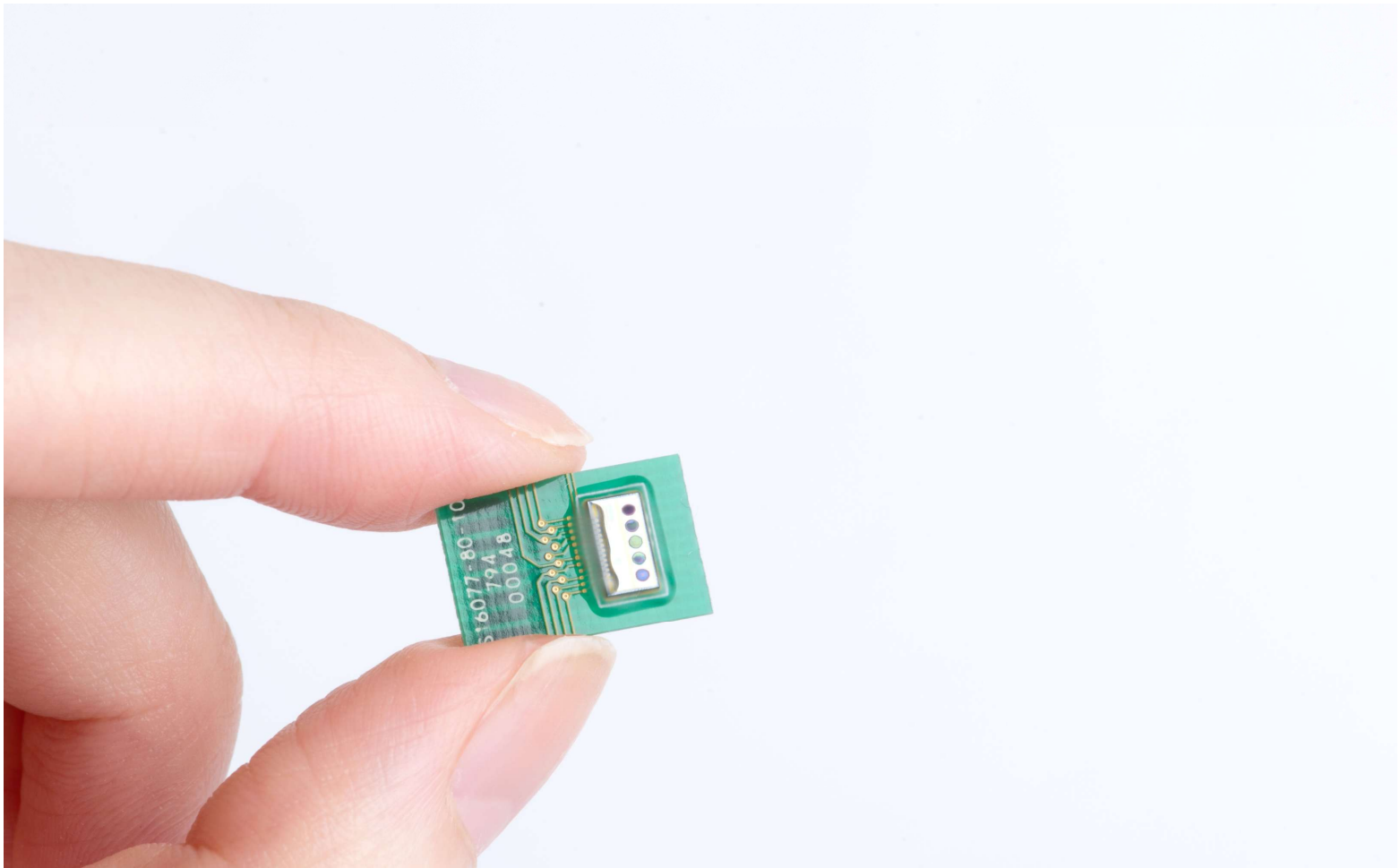
CMOS type sensor

5C-SSM

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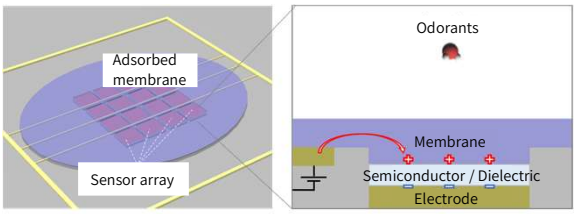


Introduction to 5C-SSM (CMOS type)

Using the CMOS sensor's ability to store an electric charge on its surface, the sensor converts the data into numerical values by detecting the increase or decrease in the amount of electric charge that occurs when smell molecules are adsorbed/desorbed on the membrane.

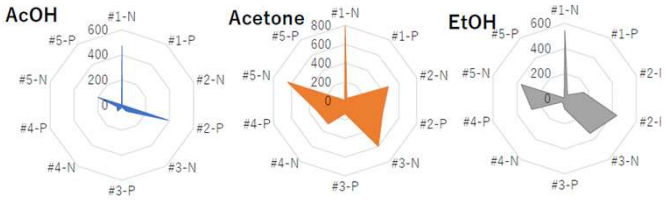
Principle of smell detection

16 sensor elements (4×4) are arranged per membrane, allowing simultaneous acquisition of output data for up to 80 elements for five different membranes. This makes it possible to avoid the risk of sensor failure and reduce noise by integrating and averaging the output data.



Example of sensor response

This is an example of output from FS0101.The output pattern differs depending on the sample. Even with the same sample, the output pattern will be different when other film sets (FS0200 / FS0300) are used.



For PoC

Membrane set: FS0101/FS0200/FS0300/Membrane Selection Custom



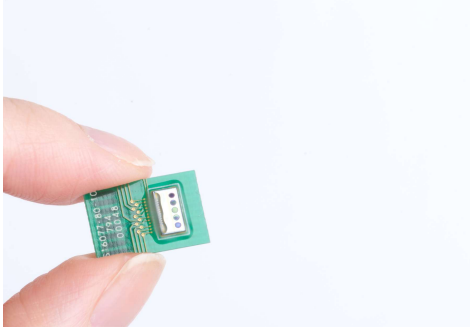
CMOS evaluation kit_box（CEK）
Evaluation kit with 5C-SSM/IF board and flow path fixture housed in a casing. The included software features automatic sensor pixel selection, enabling easy data acquisition.



5C-SSM evaluation set
Evaluation kit equipped with 5C-SSM. The interface board has obtained CE certification.

Specifications

Sensor module size：9.7×15×2mm
Sensor area size　：5.8×3.2×0.6mm
Weight　：Approx. 0.24 mg
Supply voltage　：Rated value -0.3 ~ 6.0 V at 25°C
Recommended operation temperature and humidity：
0°～50°C ＜85% *No condensation
Recommended storage temperature and humidity：
-10～60°C ＜85% *No condensation



*Note that if there is a temperature difference between the product and its surroundings in a humid environment, the surface of the product is prone to condensation, which may affect its characteristics and reliability or cause the sensor to break.

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